

Medical Email Confidentiality Annotation Guide

1. Overview

This guide is designed to assist you in the quality assurance process for the Medical Email Confidentiality Dataset. Your task is to review 2,000 bilingual (English/Arabic) emails and verify the accuracy of their assigned **Confidentiality Degree (1-5)**.

The goal is to ensure that each email is correctly classified based on the sensitivity of its content, ranging from public information to strictly confidential patient data.

2. Annotation Task Instructions

Step-by-Step Process

1. **Read the Email:** Carefully read both the English and Arabic versions of the email.
2. **Analyze Content:** Identify key information such as patient names, medical diagnoses, treatment details, internal hospital operations, or public announcements.
3. **Verify Confidentiality Degree:** Compare the assigned `confidentiality_degree` (1-5) with the criteria defined in Section 3.
4. **Check for Discrepancies:**
 - Does the English text match the Arabic translation in meaning and tone?
 - Is the assigned level appropriate for the content?
 - Are there any PII (Personally Identifiable Information) leakages in lower-level emails (Levels 1-2)?
5. **Flag Issues:** If you find a mismatch or error, note the `email_id` and the specific issue (e.g., "Incorrect Level: Should be 4, not 2" or "Translation Error: Arabic text missing patient name").

3. Confidentiality Levels (1-5)

The dataset uses a 5-level scale to categorize the sensitivity of the information. Use the following definitions and examples to guide your review.

Level 1: Public / Low Sensitivity

- **Definition:** Information intended for public release or general distribution. Contains no personal data, internal secrets, or sensitive operational details.
- **Content:** Public health announcements, press releases, published research findings (without patient data), general inquiries.
- **Example:**
 - *Subject:* "New Clinic Opening Hours"
 - *Body:* "We are pleased to announce that our clinic will now be open on Saturdays from 9 AM to 2 PM."

Level 2: Internal / Administrative

- **Definition:** Routine internal communications not intended for the public but containing low-risk information.
- **Content:** Meeting schedules, general administrative updates, marketing outreach to healthcare partners, preliminary research shared internally (no patient data).
- **Example:**
 - *Subject:* "Staff Meeting Rescheduled"
 - *Body:* "The weekly departmental meeting has been moved to Thursday at 10 AM in Conference Room B."

Level 3: Confidential / Restricted Internal

- **Definition:** Sensitive internal information that should be restricted to specific teams or departments. Disclosure could disrupt operations or violate internal policies.
- **Content:** Staffing changes, resource allocation (e.g., ICU beds, ventilators), internal policy updates regarding security or compliance, research data with moderate sensitivity.
- **Example:**
 - *Subject:* "ICU Staffing Update"
 - *Body:* "Due to increased patient load, we are reallocating two nurses to the ICU night shift effective immediately."

Level 4: Highly Confidential / Patient-Specific (Non-Critical)

- **Definition:** Private information related to specific patients, including medical history, insurance details, or treatment plans. Disclosure would violate patient privacy laws.
- **Content:** Lab results (e.g., cholesterol, blood pressure), insurance claim clarifications, appointment follow-ups with specific medical context.

- **Example:**
 - *Subject:* "Lab Results: Mr. Ahmed Al-Mansoori"
 - *Body:* "Your recent blood tests show slightly elevated cholesterol. Please schedule a follow-up to discuss dietary changes."

Level 5: Strictly Confidential / Critical

- **Definition:** Extremely sensitive information where disclosure could cause significant harm, distress, or legal/regulatory consequences.
- **Content:** Critical diagnoses (e.g., cancer, tumors, HIV), mental health records, genetic data, regulatory compliance audits identifying violations, legal disputes involving patients.
- **Example:**
 - *Subject:* "MRI Results: Confidential"
 - *Body:* "The scan revealed a lesion in the left temporal lobe. Immediate further evaluation is required to rule out malignancy."

4. Key Indicators for Quick Assessment

Level	Keywords & Indicators
1	Public, Announcement, General Info, Press Release
2	Internal, Meeting, Schedule, Marketing, Administrative
3	Restricted, Staffing, Resource Allocation, Policy Update, Security
4	Patient Name, Insurance ID, Lab Results (Routine), Appointment
5	Tumor, Lesion, Critical, Audit, Violation, Strictly Confidential

5. Tool Usage Tips

- **Bilingual Check:** Ensure the Arabic text accurately reflects the English text. Watch for mistranslations of medical terms (e.g., "Lesion" should be "آفة" or "ورم", not just

"إصابة").

- **Context Matters:** A "meeting request" (Level 2) becomes Level 4 if it mentions "discussing Mr. Smith's surgery".
 - **Consistency:** If multiple emails discuss the same patient/condition, they should generally have the same confidentiality level unless the specific content differs significantly in sensitivity.
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Note: If you are unsure about a specific email, flag it for review by the supervisor. Accuracy is critical for training our AI models.